

Resource Deployment for Travelers Insurance

A Multi-Layered Data-Driven System for Resource Allocation

Project Report for DS7900 – Applied Analytics Project Course

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Core Business Area	Catastrophe claim operations and resource deployment
Analytical Layers	Classification, on-site/virtual routing, spatial clustering, travel-time matrix, optimization, dashboard reporting
Primary Goal	Maximize CAT claim completion within SLA while reducing travel burden and deployment inefficiency

Executive Summary

This report presents a data-driven resource allocation system developed for Travelers Insurance to improve the adjuster-claim assignation process during a catastrophic event, or CAT event. During catastrophic weather events, insurance companies receive a sudden surge of claims across geographically dispersed locations. These claims must be reviewed, assigned, investigated, and closed within service-level agreement deadlines. The operational challenge is not only the number of claims, but also the complexity of matching each claim to an adjuster with the correct skill level, availability, geographic feasibility, travel ability, and work capacity.

The project addresses this challenge by creating a multi-layered response system composed of four major analytical components: a classification model, spatial clustering, an optimization model, and an operational dashboard. The classification model estimates missing CAT severity codes. The spatial clustering model groups on-site claims into geographic clusters and treats the centroids as temporary bases to reduce travel complexity. The time matrix estimates travel time and distance between claims and temporary base locations. Finally, the optimization model assigns adjusters to temporary bases and claims throughout planning horizon while respecting business constraints such as skill match, service deadlines, travel ability, daily capacity, and claim severity.

The final system showed a strong operational value. It improved claim completion under SLA, reduced unnecessary travel, reduced the number of deployed workers, and provided a structured framework for future CAT events. The final presentation reports a completion rate under SLA of 97.74 percent, travel time of 2.85 hours per claim, drive distance of 109 miles per claim, 124 workers deployed, and processing time of approximately six minutes.

Overall, the project demonstrates how predictive modeling, geographic analysis, and mathematical optimization can be combined to support better insurance operations during high-pressure catastrophic events. The system is valuable because it translates a difficult manual assignment process into a repeatable decision-support process that can be tested, monitored, and improved over time.

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1. Introduction

Catastrophe claim management is one of the most difficult operational problems in the insurance industry. When a major storm, tornado, hurricane, wildfire, or other catastrophic event occurs, thousands of policyholders may submit claims in a short period of time. These claims are often geographically dispersed, vary in severity, and require different types of adjuster expertise. Some claims can be handled virtually, while others require on-site inspection. Some claims may be low severity and can be completed quickly, while others may require highly experienced adjusters and multiple days of work.

The key challenge is that claim assignment decisions must be made quickly. Delay in assigning adjusters can increase cycle time, damage customer satisfaction, and create service-level agreement risk. At the same time, poor assignment decisions can create unnecessary travel costs, underutilized workers, overloaded adjusters, or mismatches between claim complexity and adjuster skill. The analytical plan describes this situation as a CAT event in which a major disaster creates a claims spike, strains capacity, and creates operational pressure.

Traditional claim assignment methods may rely heavily on manual judgment. While experienced managers can make good decisions, manual allocation becomes difficult when claim volume rises quickly. A manual approach can struggle to account for all relevant factors at the same time, including claim severity, adjuster skill, adjuster location, travel time, claim deadline, daily productivity, and on-site versus virtual handling. This creates the need for a scalable, data-driven decision framework.

The purpose of this project is to design and evaluate such a framework. The project uses Travelers Insurance claim and adjuster data to build a decision-support system for resource allocation. Instead of treating claim assignments as a single process, the project breaks the problem into multiple analytical layers. First, the data is cleaned and standardized. Second, missing claim severity values are predicted using a classification model. Third, claims are separated into virtual and on-site handling categories. Fourth, on-site claims are grouped geographically using spatial clustering. Fifth, travel time and distance are estimated. Finally, an optimization model assigns adjusters to claims and clusters while maximizing claim completion and controlling operational cost.

This report explains the project in the style of an academic paper. It covers the business problem, project goals, expected outcomes, data summary, target variables, data preparation, model selection, model usage, results, limitations, recommendations, and conclusions. The report has been written to explain not only what was built, but also why each analytical method was selected and how each method contributes to the final business solution.

2. Business Problem and Business Impact

The main business problem is the lack of a systematic, scalable method for matching adjusters to incoming CAT claims. There is currently no systematic approach to efficiently match adjusters to incoming claims. The strategic direction is to shift from a manual assignment

process to a data-driven process with systematic matching, maximized SLA completions, and minimized cost.

This business problem has several operational consequences. The analytical plan identifies major consequences of inefficient allocation, including skill mismatch, mobility logistics, capacity imbalance, time-driven claim exposure, service-level breach, and customer experience erosion. Each consequence matters because CAT events are high-volume, high-pressure situations where small assignment errors can become large operational problems.

2.1 Skill Mismatch

Skill mismatch occurs when an adjuster is assigned to a claim that requires a different level or type of expertise. For example, a low-skill adjuster may be assigned to a high-severity claim, or an adjuster with personal lines of expertise may be assigned to a business lines claim. This can slow down resolution, reduce claim quality, and create rework. In CAT operations, rework is especially costly because the claim volume is already high. A mismatch can also cause customer frustration if the adjuster cannot answer questions or move the claim forward efficiently.

2.2 Mobility Logistics

Mobility logistics are another major issue. On-site claims require adjusters to travel. If adjusters are assigned without considering location of claims, temporary bases, or route feasibility, the company may spend more time and money on travel than necessary. Excessive travel also reduces the time adjusters can spend actually resolving claims. This creates a direct trade-off between travel burden and productive claim handling time. The project therefore treats travel not as a secondary issue, but as a central operational constraint.

2.3 Capacity Imbalance

Capacity imbalance occurs when some adjusters are overloaded while others are underutilized. During CAT events, capacity must be managed carefully because daily claim volume can exceed normal operating conditions. Without a system, certain locations or claim types may develop bottlenecks. An overloaded group of adjusters may miss deadlines, while another group may have unused capacity. A good assignment system should distribute work according to skill, capacity, and demand rather than relying on informal allocation.

2.4 Time-Driven Claim Exposure

Time-driven claim exposure refers to the risk that claims become more costly or difficult when they are not addressed promptly. Delayed inspections and delayed communication may increase customer frustration and extend the settlement cycle. For severe claims, delays can also create downstream costs because damaged property may deteriorate further, or customers may require additional support. The SLA deadline is therefore not just an administrative rule; it represents a time-sensitive service commitment.

2.5 Service-Level Breach and Customer Experience

Service-level breach is especially important because CAT claims often have deadlines based on severity. Missing those deadlines can affect compliance, customer trust, and operational performance. Customer experience also suffers when policyholders wait too long during an already stressful time. A policyholder filing a CAT claim is often dealing with property damage, displacement, or financial stress. Faster and more accurate assignment improves customer service and helps protect brand reputation.

The project is therefore not only a technical exercise. It is designed to improve real business outcomes. By assigning the right adjusters to the right claims at the right time, the company can increase SLA completion, reduce operating costs, reduce unnecessary travel, and improve customer experience. The business impact is strongest because the final solution links operational rules, predictive modeling, geographic planning, and optimization into one repeatable process.

3. Project Goal and Expected Outcomes

The main goal of the project is to develop a scalable decision-support system for CAT claim allocation. The business goal is to establish a scalable decision framework that optimizes adjuster allocation and routing to control total operating cost while maintaining service performance standards and customer satisfaction during peak demand periods.

There are several outcomes expected from the project: First, the system should identify which adjusters are available and qualified. This requires cleaning adjuster data and filtering the adjusters with 0 skill level in both types of expertise. Availability is a basic requirement because inactive or unavailable adjusters should not be considered in the assignment pool. Qualification is equally important because claim complexity and division must match adjuster capability.

Second, the system should identify which claims need to be solved. This requires cleaning claim records, identifying the event window, assigning severity, calculating due dates, and separating claims into on-site or virtual handling. Claims are not identical. A severity 5 claim with a seven-day deadline is very different from a severity 1 claim with a twenty-eight-day deadline. Therefore, the system needs a structured claim profile for each claim.

Third, the system should determine whether a specific adjuster can work on a specific claim. This requires business rules around skill level, division, travel ability, and claim handling mode. The model must validate location, match geography, and align adjuster expertise with claim division and severity. These rules prevent infeasible or inappropriate assignments.

Fourth, the system should estimate how much work is possible. This requires a productivity matrix that converts claim severity and adjuster skill into service days or claim handling capacity. The productivity matrix is essential because a model cannot assign work intelligently unless it knows how much work each claim requires and how much capacity each adjuster can provide.

Fifth, the system should produce a final optimized assignment. The output should show which adjusters are assigned to which claims or clusters, how many claims are completed within SLA, how much travel is required, and how many workers are deployed. The assignment should maximize completion under constraints rather than simply assigning claims as fast as possible.

Finally, the system should support decision-making through dashboard reporting. The analytical solution should be delivered as a decision-support dashboard powered by an optimization model that maintains business rules and assumptions. This dashboard layer is important because managers need interpretable outputs, not just model files. The expected outcome is a practical tool that can support real CAT operations after additional testing and validation.

4. Data Summary and Target Variables

The project used two main operational datasets: claim data and adjuster data. The claim dataset contained CAT claim records, including claim number, notice of loss date, division, severity code, accident ZIP code, peril description, weather text, cause of injury text, and other claim-related fields. The adjuster dataset contained roster information, including adjuster ID, skill levels, travel eligibility, location, and availability-related fields.

The analytical plan reports 1,065 total claims and 774 total adjusters for the CAT event window from December 9 to December 30. In the cleaned modeling files, the claims were further processed, and the final optimization model used 1,053 feasible claims after removing structurally impossible claims. The adjuster records were grouped into 19 resource groups based on skill, travel ability, and eligible cluster signature. Grouping resources helped reduce the size of the optimization problem while preserving the key operational differences among adjuster types.

Data Element	Description
Claims data	CAT claim records including claim number, notice of loss date, division, severity, location, and descriptive text fields.
Adjuster data	Roster data including adjuster ID, skill levels, willingness to travel, location, and availability-related information.
Productivity matrix	Business-rule table connecting claim severity and adjuster skill level to service days or productivity rates.
Travel matrix	Estimated travel time and distance between adjuster groups, claim locations, and claim clusters.
Dashboard outputs	Business-facing metrics such as SLA completion, travel burden, deployment count, and processing time.

4.1 Target Variables

The project had two major target variables. The first target variable was *CAT Severity Code*. This was the target for the classification model. The CAT Severity Code represents the severity or complexity of a claim. Severity is important because it influences SLA days, productivity, adjuster skill requirements, and priority. However, the project recognized that severity codes may be missing or imperfect. Therefore, *CAT Severity Code* was identified as a data challenge because initial phone triage may produce inaccurate assessments.

The second target outcome was claim completion within SLA. This was the main target of the optimization model. The optimization model was designed to maximize the number of completed claims while considering weighted severity, due dates, productivity, capacity, and travel. In practical terms, the model asks how the organization can assign adjusters so that the largest possible number of claims are completed before their deadlines.

4.2 Key Variables

Other important variables included *Claim Number*, *NOL Date*, *Due Date*, *Division*, *Accident ZIP*, *latitude and longitude*, *on-site or virtual handling status*, *CL Skill Level*, *PL Skill Level*, *Will Travel*, *service days*, and *cluster ID*. Each variable had a specific role. *Claim Number* identified claims. *NOL Date* defined when claims entered the system. *Due Date* created the SLA window. *Division* determined whether CL or PL skill should be used. *Accident ZIP* and coordinates made geographic analysis possible. Skill levels and travel flags determined adjuster eligibility. *Service days* made capacity modeling possible. *Cluster ID* connected on-site claims to geographic deployment areas.

Together, these variables allowed the project to move from raw operational records to a structured optimization problem. This is a key strength of the project. The project did not simply run models on raw data. It transformed business records into decision-ready variables that could support operational planning.

5. Data Preparation

Data preparation was one of the most important parts of the project because the final optimization model depends on clean, standardized, and logically consistent input data. Poor data quality would lead to poor assignments, invalid travel estimates, or infeasible optimization results. The data preparation process included claim cleaning, adjuster cleaning, severity imputation, productivity matrix construction, geographic preparation, and travel time matrix creation.

5.1 Claim Data Cleaning

The claim cleaning process began by selecting the most important fields from the claim dataset. These included Claim Number, NOL Date, Division, CAT Severity Code, and Accident ZIP. The CAT Severity Code was converted into a numeric field, and rows with missing or invalid essential values were removed or corrected where possible. The cleaning process also checked ZIP code quality because accident ZIP code was necessary for geocoding and travel estimation.

Then service-level agreement days were assigned based on severity. Higher-severity claims were assigned shorter deadlines, while lower-severity claims received longer deadlines. The business rule used regarding Service Level Agreement is as follows:

Severity	Service Level Agreement
1	28 days
2	28 days
3	21 days
4	14 days
5	7 days

The due date was calculated as NOL Date plus SLA days. This due date became a key constraint in the optimization model because claims had to be completed within their SLA window.

Next, claims were divided into on-site and virtual handling categories. This was important because virtual claims do not require the same travel planning as on-site claims. To achieve this, automated daily assignment of incoming CAT claims to virtual handling and in-person handling was done based on severity. This separation improved the realism of the model because field work and virtual work have different resource requirements.

The project also cleaned ZIP codes and converted them into latitude and longitude. This step allowed claims to be mapped geographically and included in clustering and travel calculations. Rows with invalid or missing ZIP information were identified. In the cleaned data, the majority of claim ZIP codes were usable, while only a very small number had missing ZIP values. This high level of usable location data made the spatial modeling layer possible.

5.2 Adjuster Data Cleaning

The adjuster dataset required no cleaning process; but instead required filtering to establish an eligible pool. The fields used were the adjuster id, their skill level of either personal or business division, their availability to travel and work. We filtered all adjusters that had a skill level of 0 in both divisions and the ones that were not available to work.

5.3 Productivity Matrix Preparation

The productivity matrix converted claim severity and adjuster skill into expected service days. This was necessary because different adjusters can complete different claims at different speeds. A highly skilled adjuster can complete lower-severity claims quickly, while a lower-skilled adjuster may be unable to handle high-severity claims. Without this productivity matrix, the optimization model would not know how much work each claim requires.

The matrix used CAT Severity Code and Skill Level as inputs and produced service days as output. For example, a high-severity claim handled by a high-skill adjuster may require several service days, while lower-severity claims may be completed faster. The productivity matrix translated business knowledge into mathematical constraints.

This matrix was critical because it created a bridge between business rules and optimization. It allowed the model to represent work in terms of capacity. If a group of adjusters has a certain number of available workdays, and each claim requires a certain number of service days, then the model can calculate whether the group has enough capacity to complete its assigned claims before the deadline.

5.4 Geographic Preparation

For on-site claims, the zip codes were converted to latitude and longitude and then into projected X and Y coordinates using a projected coordinate reference system. This was necessary because K-Means clustering works better with distance in projected units such as meters rather than raw latitude and longitude degrees. Latitude and longitude are angular measurements, while clustering requires a meaningful distance scale.

After projection, claims were ready for clustering. Clustering reduced the geographic complexity of the problem by grouping nearby claims together. Instead of considering every on-site claim independently for deployment, the system could assign resources to clusters and reduce travel complexity. This preparation step directly supported the spatial clustering and optimization layers.

5.5 Quality Control During Preparation

Several quality-control checks were performed during preparation. ZIP code audits identified missing or invalid accident ZIP codes. Geocoding checks confirmed whether latitude and longitude were available for claims and adjusters. Feasibility checks in the optimization model removed structurally impossible claims, meaning claims that could not be completed under the available business rules and resource structure. These checks improved reliability because infeasible data can cause misleading model results.

6. Model Selection and Model Usage

This project presents several models and analytical methods. Each model was selected for a specific purpose. The system did not rely on a single model; instead, it used a pipeline of models, where each model solved one part of the operational problem. This layered design is one of the strongest features of the project because it reflects the real-world complexity of CAT resource allocation.

6.1 Classification Model for CAT Severity Imputation

Why the Model Was Needed

The classification model was used to predict missing CAT Severity Codes. Severity is a key variable because it affects SLA days, required skill level, productivity, and claim priority. If

severity is missing, the claim cannot be properly included in downstream assignment and optimization logic. Therefore, the first modeling problem was to impute missing severity values using available claim information.

Rows with missing severity were processed through the same text preprocessing pipeline and TF-IDF features before being predicted using the trained SVM model. In this way we used a consistent transformation process for both training data and new missing-severity rows.

Why LinearSVR Was Picked

The final classification model used a Linear Support Vector Regression approach, specifically LinearSVR, even though the target was ordinal severity from 1 to 5. This choice makes sense because *CAT Severity Code* is not simply a nominal class. Severity levels have order: 5 is more severe than 4, 4 is more severe than 3, and so on. Treating severity as an ordinal numeric outcome allows the model to learn the relative distance between categories.

LinearSVR was also appropriate because the model used high-dimensional text features generated by TF-IDF. Text data often creates many sparse features, especially when using word n-grams. Linear models tend to perform well on sparse text data because they are computationally efficient and can handle many predictors. A more complex nonlinear model may overfit, especially when the labeled dataset is relatively small.

The model used *Cause of Injury Text*, *Division*, *Weather Text*, *Peril Description*, *Peril Group*, TF-IDF word features, semantic rule-based tags, and one-hot encoded categorical variables. The TF-IDF component allowed the model to capture important words and phrases from the claim description. The categorical features added structured claim context. The semantic tags added rule-based domain signals, such as whether the text suggested roof damage, water damage, injury, or other meaningful claim characteristics.

How the Model Performed

The classification model went through several experiments. The final selected model, Model L with optimized thresholds, performed better than simple naive rounding. The final metrics showed naive rounding accuracy of 45.26 percent, naive rounding MAE of 0.7067, and naive rounding QWK of 0.4864. With optimized thresholds, accuracy improved to 47.66 percent, MAE decreased to 0.6903, and QWK improved to 0.5477.

Accuracy improved slightly, MAE decreased, and Quadratic Weighted Kappa improved meaningfully. QWK is especially important for ordinal classification because it rewards predictions that are close to the true class and penalizes large severity mistakes more heavily. For example, predicting severity 4 when the true value is 5 is less serious than predicting severity 1 when the true value is 5.

The model did not achieve very high accuracy, but the performance was reasonable given the limitations. We note that *Cause of Injury Text* alone had limited predictive power, the dataset was small, and class imbalance negatively affected performance. Therefore, the model should be judged as useful for imputation support, not as a perfect automated severity engine. It strengthened downstream modeling by reducing missingness in a key field.

Severity Model Version	Accuracy	MAE	QWK
Naive rounding	45.26%	0.7067	0.4864
Optimized thresholds	47.66%	0.6903	0.5477
Interpretation	Improved	Lower error	Better ordinal agreement

6.2 On-Site and Virtual Handling Logic

Why This Step Was Needed

Not all claims require the same handling process. Some claims can be handled virtually, while others require an adjuster to physically inspect the damage. This distinction is important because on-site claims create travel requirements, while virtual claims primarily require skill and capacity. Treating every claim as on-site would overstate travel needs, while treating every claim as virtual would ignore field inspection requirements.

Why It Was Picked

The project used a standardized routing logic to classify claims into virtual handling and in-person handling. This logic was picked because it mirrors real-world operations. Claims arrive daily, and triage decisions are often made based on available claim information. By processing claims in daily batches by NOL Date, the project mimicked the actual daily intake workflow. This approach supports operational surge planning, prevents cross-day allocation bias, and enables independent daily decision-making.

How It Performed

The handling logic helped split the modeling problem into two separate operational channels. Virtual claims could be handled without travel, while on-site claims moved into the spatial clustering and travel matrix process. This made the optimization problem more realistic and easier to solve. It also improved dashboard interpretation because managers could separately monitor virtual workload and field workload.

6.3 Adjuster Filtering and Skill Matching

Why This Step Was Needed

A claim should only be assigned to an adjuster who is qualified to work on it. The adjuster filtering process determined how many adjusters were qualified to handle each CAT claim based on availability, skill level, claim division, and minimum severity requirements.

Why It Was Picked

This rule-based filtering was picked because claim assignment is constrained by business logic. A machine learning model alone cannot decide whether an adjuster is operationally or professionally allowed to handle a claim. The model must respect business rules. The filtering logic translated severity into minimum required skill, removed unavailable adjusters, removed

trainees or adjusters with skill level zero, used CL Skill Level for business-related claims, used PL Skill Level for property-related claims, and counted eligible adjusters for each claim.

How It Performed

The filtering process produced required skill level and number of eligible adjusters for each claim. This was valuable because it showed whether the company had enough qualified resources for different types of claims. It also reduced the optimization search space by removing infeasible claim-adjuster combinations before optimization. Operationally, this helps detect skill mismatches, identify high-severity capacity gaps, quantify surge strain, and support workforce planning decisions.

6.4 K-Means Spatial Clustering

Why Clustering Was Needed

On-site claim assignment requires travel planning. If every on-site claim is treated as an independent location, the optimization model becomes more complicated and may create inefficient deployment patterns. Clustering solves this problem by grouping nearby claims into geographic regions. The purpose was to create compact claim clusters so that adjusters could be deployed more efficiently to geographic areas.

Why K-Means Was Picked

K-Means was picked because it is simple, efficient, and well suited for grouping points based on distance. The goal was not to discover hidden customer segments; the goal was to create operational territories for field deployment. K-Means is appropriate for this because it minimizes distance between points and their assigned cluster centroids. It also scales well and produces cluster centroids that can be used as representative locations for travel analysis.

The model evaluated K values from 2 to 30. The selection was based on average distance to centroid, maximum distance to centroid, and percentage improvement per additional cluster. The project used a 2 percent improvement threshold to identify the point where adding more clusters no longer produced enough benefit. This is similar to an elbow method, but with an explicit operational threshold.

How It Performed

The clustering analysis selected K = 19. At K = 19, the average distance to centroid was approximately 34,388.87 meters, and the maximum distance to centroid was approximately 306,725.59 meters. The uploaded distance analysis chart showed that distance decreased as K increased, but improvement became smaller after K = 19. Therefore, K = 19 was chosen as a practical balance between compactness and complexity.

This was a reasonable modeling decision. A smaller number of clusters would create larger territories and more travel burden. A much larger number of clusters would reduce travel distance but increase model complexity and deployment fragmentation. K = 19 provided a middle ground. It allowed the field assignment model to operate at a manageable geographic level while still preserving enough location detail to reduce unnecessary travel.

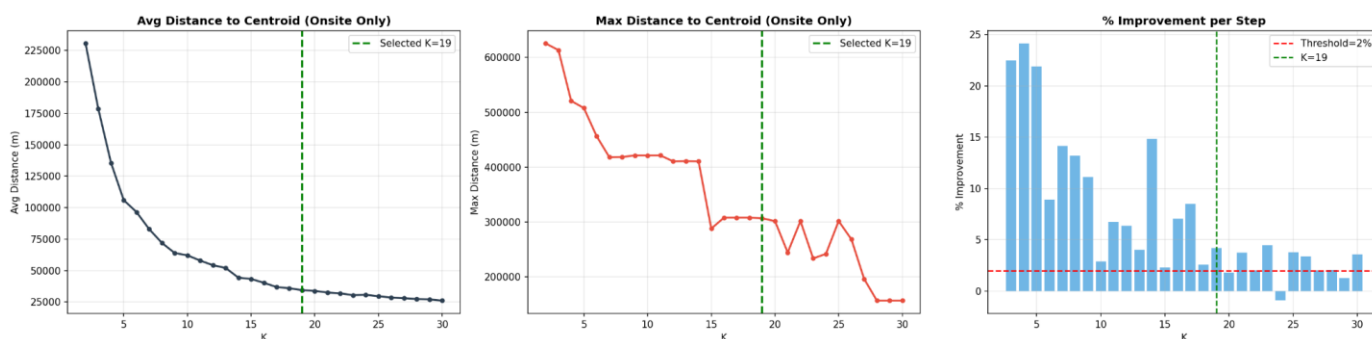


Figure 1. Distance analysis used to select K = 19 for on-site claim clustering.

6.5 Time Matrix and Travel Estimation

Why the Time Matrix Was Needed

Travel time is one of the most important cost drivers in on-site CAT claim operations. Even if an adjuster is skilled enough to handle a claim, the assignment may be inefficient if the adjuster must travel too far. The time matrix estimates travel time and distance between adjusters and claim clusters or claim locations.

Why This Method Was Picked

We used routing-based travel estimates, including OSRM-style travel logic, because straight-line distance is not accurate enough for operational planning. Real travel depends on roads, geography, and driving distance. A routing-based estimate gives more useful travel hours and miles. This matters because a model that only uses straight-line distance can understate the burden of rural routes, highway access, or geographic barriers.

How It Performed

The time matrix allowed the optimization model to include deployment and travel penalties. It also allowed the final result to report operational metrics such as total travel hours and total drive miles. The optimized model achieved a lower travel burden than the greedy baseline. In the notebook results, optimized travel hours were 3,029.16, and optimized drive miles were 116,523.25.

6.6 Gurobi Mixed-Integer Programming Optimization Model

Why Optimization Was Needed

The final assignment problem is too complex for simple manual decision-making. Each claim has a due date, severity, division, location, and required skill. Each adjuster has skill levels, travel eligibility, location, and capacity. Some claims require on-site deployment, while others can be handled virtually. The model must decide which resources should work on which claims and when.

A greedy approach can make fast assignments, but it may not find the best global solution. Greedy methods often assign the best immediate match without considering future capacity or downstream constraints. During CAT events, this can create bottlenecks and missed SLA

opportunities. Therefore, the project used a mathematical optimization model solved with Gurobi.

Why Gurobi Was Picked

Gurobi was picked because the assignment problem is a mixed-integer optimization problem. The model includes binary and integer decision variables, capacity constraints, deadline constraints, deployment constraints, and objective trade-offs. Gurobi is designed for large-scale optimization problems of this type. It can solve problems where decisions must be integer-valued, such as whether a claim is completed or how many adjusters from a group are deployed.

The model included decision variables such as $x[g,c]$, representing the number of adjusters from group g deployed to cluster c ; $w[g,i,d]$, representing the number of adjusters from group g working claim i on day d ; and $z[i]$, representing whether claim i is completed. These variables allowed the model to represent both deployment and daily claim work.

The objective was to maximize weighted claim completion while penalizing unnecessary deployment and travel. This objective aligns with the business goal: complete claims within SLA while controlling cost. The weighted completion component ensures that more important or severe claims receive appropriate priority, while penalties prevent the model from deploying too many workers unnecessarily.

How the Model Performed

The optimization model performed strongly, resulting in

- 1,038 of 1,053 claims completed within SLA, representing a 98.58% completion rate
- Weighted completion of 97.94 %
- Average cycle time was 14.90 days, median cycle time was 14.00 days
- Adjuster utilization of 6.57%
- Total travel hours = 3,029.16 (2.85 hours per claim)
- 116,523.25 total drive miles (109 miles per claim)
- Deployment count of 124 adjusters
- Model processing runtime of 342.58 seconds (~ 6 minutes)

The model also passed several quality assurance checks. These checks confirmed that the solution existed, work windows were respected, deployment counts did not exceed group size, group-day capacity was respected, completion logic was correct, and overall completion rate was strong. These checks are important because an optimization model can sometimes produce mathematically valid but operationally unrealistic outputs if constraints are missing or incorrectly defined. The checks increased confidence that the model reflected the intended business rules.

7. Results and Performance Evaluation

The results show that the project successfully created an end-to-end decision-support system. The final performance can be judged at three levels: severity prediction, spatial clustering, and assignment optimization.

7.1 Classification Results

The classification model improved severity imputation. The optimized threshold version achieved better ordinal agreement than naive rounding. Although accuracy was not extremely high, the model was useful because it filled missing severity values and allowed downstream modeling to continue. The main lesson from the classification model is that text alone is not enough to perfectly predict severity. Cause of injury text had limited predictive power, the small dataset restricted the model's ability to learn complex patterns, and class imbalance hurt performance.

Even with those limitations, the model added value. In a business environment, a moderate-quality imputation model can be useful if it is combined with human review, confidence scoring, and later operational validation. The model should not replace claim triage expertise, but it can reduce missing data and support faster planning.

7.2 Clustering Results

The clustering model selected 19 on-site clusters. This helped reduce geographic complexity while keeping average distance to centroids at a reasonable level. The selected K balanced operational simplicity and travel efficiency. The cluster layer also made the optimization model easier to solve because resources could be deployed to a limited number of geographic groups rather than every claim location independently.

7.3 Optimization Results

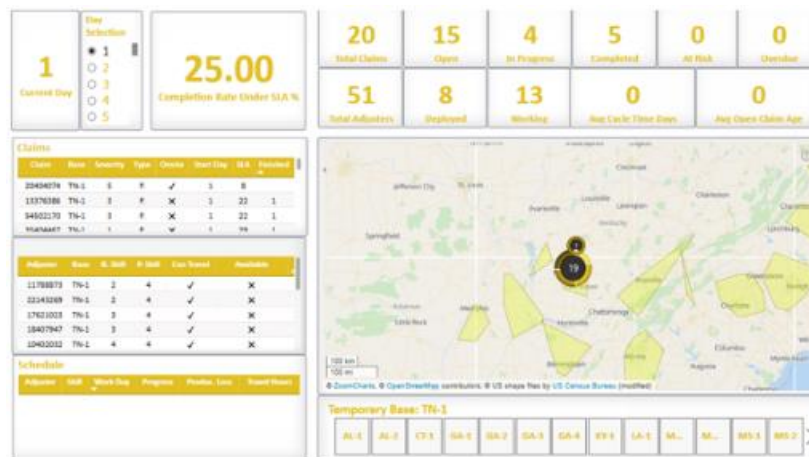
Metric	Optimization Model
Claims completed within SLA	1,038 / 1,053
Completion rate	98.58%
Weighted completion	97.94%
Travel hours	3,029.16
Drive miles	116,523.25
Deployments	124

The optimization model was the strongest part of the project. These results show that the optimization model did not simply complete more claims by using more resources. Instead, it completed more claims while using fewer deployments and less travel. That is the strongest

evidence that the model created real operational value. A model that maximizes SLA completion but dramatically increases travel or deployment would be less compelling. This model improved both service performance and operational efficiency.

7.4 Business-Facing Dashboard Results

The final dashboard output translates technical outputs into business-facing performance indicators. These included completion rate under SLA, travel time per claim, drive distance per claim, number of workers deployed, and system processing time. These metrics are important because executives and operations managers need clear indicators rather than raw optimization variables. The dashboard framing makes the model easier to adopt because it communicates results in terms of service, cost, and decision speed.



8. Discussion

The project connects technical modeling to a clear business problem. Each model had a specific operational role. The classification model solved the missing severity problem. The handling logic separated virtual and on-site claims. The adjuster filtering logic enforced business rules. The productivity matrix translated skill and severity into workload. The clustering model reduced geographic complexity. The time matrix measured travel burden. The optimization model made the final assignment decision. The dashboard communicated the result to managers.

This layered design is important because no single model could solve the entire problem. A classification model alone would not assign adjusters. A clustering model alone would not respect SLA deadlines. A dashboard alone would not optimize work. The value comes from combining all components into one decision-support pipeline.

Another strength is that the project balances predictive modeling with prescriptive analytics. Predictive modeling answers what the severity is likely to be. Prescriptive analytics answers what action the organization should take. In many business projects, predictive models are built but not connected to decisions. Here, the severity model feeds into SLA calculations, eligibility rules, productivity assumptions, and the optimization model. This makes the prediction operationally useful.

The project also uses interpretable business constraints. Skill requirements, travel ability, service days, due dates, and capacity are all understandable to managers. This is important because operational leaders are more likely to trust a model when they can understand the logic behind recommendations. The model does not act like a black box; it uses clear rules and measurable inputs.

9. Limitations

Despite the strong results, the project has several limitations. These limitations should not be viewed as failures. Instead, they identify areas for future improvement before the system is used in live CAT operations.

9.1 Classification Data Limitations

First, the classification model was limited by data quality. The cause of injury text was unstructured and not always detailed enough to predict severity accurately. The dataset was also relatively small for a text classification problem, with approximately 791 labeled records used in the classification experiments. Class imbalance also affected performance. Some severity categories had fewer examples, making it harder for the model to learn reliable patterns.

Second, the severity code itself may not be a perfect ground truth. Initial phone triage may produce inaccurate severity assessments. If the original labels are noisy, then the model can only learn an imperfect representation of severity. This means the model's performance is limited not only by algorithm choice but also by label quality.

9.2 Productivity and Capacity Assumptions

The productivity matrix was based on assumptions. While the assumptions were reasonable, actual productivity may vary by adjuster, claim type, documentation requirements, weather conditions, customer availability, and regional conditions. Some adjusters may be faster or slower than their skill level suggests. Some claims may require more work than their severity code indicates. Future versions should validate the productivity matrix using historical claim closure data.

9.3 Travel Estimate Limitations

Travel estimates may not perfectly match real conditions. Road closures, traffic, storm damage, fuel availability, lodging constraints, and local access restrictions can affect travel during CAT events. A routing engine can estimate normal drive time, but CAT events often create abnormal travel conditions. Therefore, travel estimates should be updated with real-time information when possible.

9.4 Grouping and Individual-Level Differences

The model grouped adjusters into resource groups. This improves computational efficiency, but it may hide individual-level differences such as experience, workload preference, customer service quality, local knowledge, or current fatigue. Grouping is practical for optimization, but future versions may need a hybrid approach that uses grouping for planning and individual-level assignments for execution.

9.5 Objective Function Scope

The model focused heavily on completion, travel, and deployment. It did not fully include fairness, employee fatigue, lodging cost, overtime cost, claim value, customer vulnerability, or regulatory priority. These factors can matter in real business operations. For example, two assignments may look equal in terms of SLA completion but differ in cost, employee impact, or customer sensitivity.

9.6 Validation Scope

The model was tested on one historical event window. More historical CAT events should be used to confirm that the model generalizes well. Different events may have different geographic patterns, claim severity distributions, adjuster availability, and travel constraints. A model that performs well on one event should be stress-tested before deployment.

10. Recommendations

The first recommendation is to continue using the optimization-based assignment model as the core decision-support tool. The optimization model should be treated as the central engine of the multi-layered response system.

The second recommendation is to test the model on additional historical CAT events, by conducting tests with historical deployment and event data before using the system with real CAT events. This would help validate performance across different storm types, regions, claim volumes, and adjuster rosters. Historical back-testing would also help estimate how often the model improves outcomes and under what conditions it struggles.

The third recommendation is to improve the severity classification model. More labeled data should be collected, and additional features should be added, such as claim notes, property type, damage description, policy form, loss amount estimates, photos, and inspection outcomes. Because the current model struggled with limited text and class imbalance, future versions should consider ordinal classification models, tree-based models, transformer-based text embeddings, or cost-sensitive learning.

The fourth recommendation is to strengthen the productivity matrix using real historical completion data. Instead of relying only on assumed productivity rates, the company should estimate service days from historical claim closure patterns by severity, division, adjuster skill, region, and claim type. This would make the optimization model more realistic and more defensible.

The fifth recommendation is to integrate real-time availability. Adjuster availability can change due to illness, travel delays, workload, lodging, or reassignment. A real-time dashboard should update capacity and re-run the optimization model as conditions change. The system should not be a one-time static plan; it should support continuous monitoring and re-optimization.

The sixth recommendation is to include additional cost components. Future models should include lodging cost, overtime cost, mileage reimbursement, travel fatigue, claim value, and customer priority. This would make the objective function more realistic. It would also allow

leadership to compare different strategies, such as maximizing SLA completion versus minimizing cost while maintaining a required SLA threshold.

The seventh recommendation is to use the dashboard as a managerial decision tool, not a fully automatic replacement for human judgment. The model should provide recommended assignments, SLA risk alerts, travel summaries, and capacity gaps, but managers should be able to adjust assignments based on context. Human review is especially important for unusual claims, high-value claims, customer-sensitive cases, and emergency situations.

In short, adapting the system to day-to-day operations, conducting historical CAT event tests, including real-time monitoring metrics, and implementing the system for real CAT events.

11. Conclusion

This project developed a complete data-driven system for CAT claim resource allocation at Travelers Insurance. The project began with a real business problem: during catastrophic events, claims surge rapidly, and manual assignment methods can lead to inefficient resource allocation, skill mismatches, unnecessary travel, SLA misses, and poor customer experience.

To solve this problem, we built a multi-layered analytical pipeline. Claim and adjuster data were cleaned and standardized. Missing CAT severity codes were estimated using a LinearSVR classification model with TF-IDF text features, categorical variables, and optimized thresholds. Claims were divided into virtual and on-site handling groups. Adjusters were filtered based on availability, skill level, division, and travel ability. A productivity matrix translated claim severity and adjuster skill into service days. On-site claims were grouped into 19 spatial clusters using K-Means. A travel-time matrix estimated drive time and mileage. Finally, a Gurobi optimization model assigned adjusters to claims while maximizing SLA completion and controlling deployment and travel burden.

The final system performed strongly. It completed approximately 98.58 percent of feasible claims within SLA. It also reduced travel hours, drive miles, and deployment count. The final results show business-facing metrics of 97.74 percent completion under SLA, 2.85 travel hours per claim, 109 drive miles per claim, 124 workers deployed, and six minutes of processing time.

The project's most important contribution is that it turns a complex operational challenge into a structured decision-support system. It does not rely on one model alone. Instead, it combines classification, routing rules, skill matching, spatial clustering, productivity modeling, travel estimation, optimization, and dashboard reporting. This makes the solution practical, explainable, and aligned with business needs.